Algorithms 2

Minimum Set-Cover

Daniel Rosenberg & Michael Trushkin

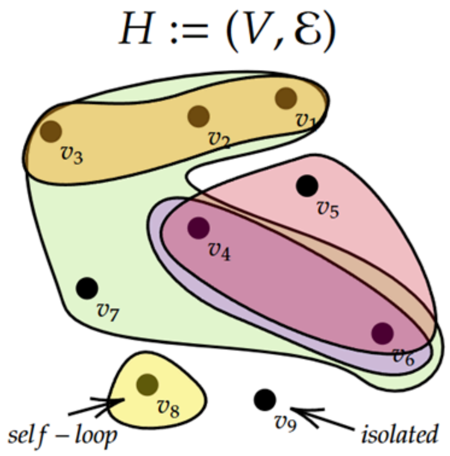
Ariel University

1. The Minimum Set-Cover Problem

Definition 1 Hypergraph

By a **hypergraph** we mean an ordered pair $H := (V, \mathcal{E})$ with $\mathcal{E} \subseteq \mathcal{P}(V)$.

- If all members of $\mathcal{E}$ are **pairs**, then H is called a **graph**
- Members of $\mathcal{E}$ are called **hyper-edges**



$V = \{v_1, \ldots, v_9\}$
$\mathcal{E} = \{e_1, e_2, e_3, e_4, e_5\}$
$e_1 = \{v_1, v_2, v_3\}$
$e_2 = \{v_1, v_2, v_3, v_4, v_5\}$
$e_3 = \{v_4, v_6\}$
$e_4 = \{v_4, v_5, v_6\}$
$e_5 = \{v_8\}$

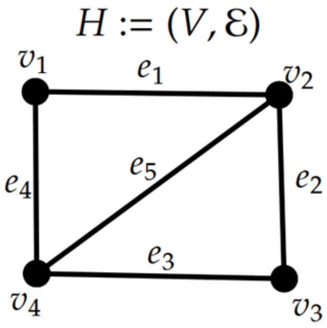
Definition 2 Hypergraph

By a **hypergraph** we mean an ordered pair $H := (V, \mathcal{E})$ with $\mathcal{E} \subseteq \mathcal{P}(V)$.

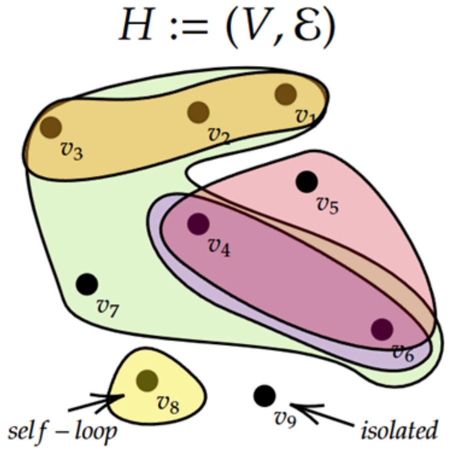
- If all members of $\mathcal{E}$ are **pairs**, then H is called a **graph**
- Members of $\mathcal{E}$ are called **hyper-edges**

FYI: If $|e| = k$ for all $e \in \mathcal{E}$, then H is called ***k*-uniform**.

- A graph is a **2-uniform** hypergraph.



$V = \{v_1, v_2, v_3, v_4\}$
$\mathcal{E} = \{e_1, e_2, e_3, e_4, e_5\}$
$e_1 = \{v_1, v_2\}$
...



$V = \{v_1, \ldots, v_9\}$
$\mathcal{E} = \{e_1, e_2, e_3, e_4, e_5\}$
$e_1 = \{v_1, v_2, v_3\}$
$e_2 = \{v_1, v_2, v_3, v_4, v_5\}$
$e_3 = \{v_4, v_6\}$
$e_4 = \{v_4, v_5, v_6\}$
$e_5 = \{v_8\}$

Problem 3 Minimum Set-Cover (SC)

Instance: A hypergraph $H := (V, \mathcal{E})$.

Goal: A subset $C \subseteq \mathcal{E}$ such that $\bigcup_{e \in C} e = V$, minimising $|C|$.

Put another way: We seek a minimum **hyper-edge-cover** of H.

1.2 The Problem

Problem 4 Minimum Set-Cover (SC)

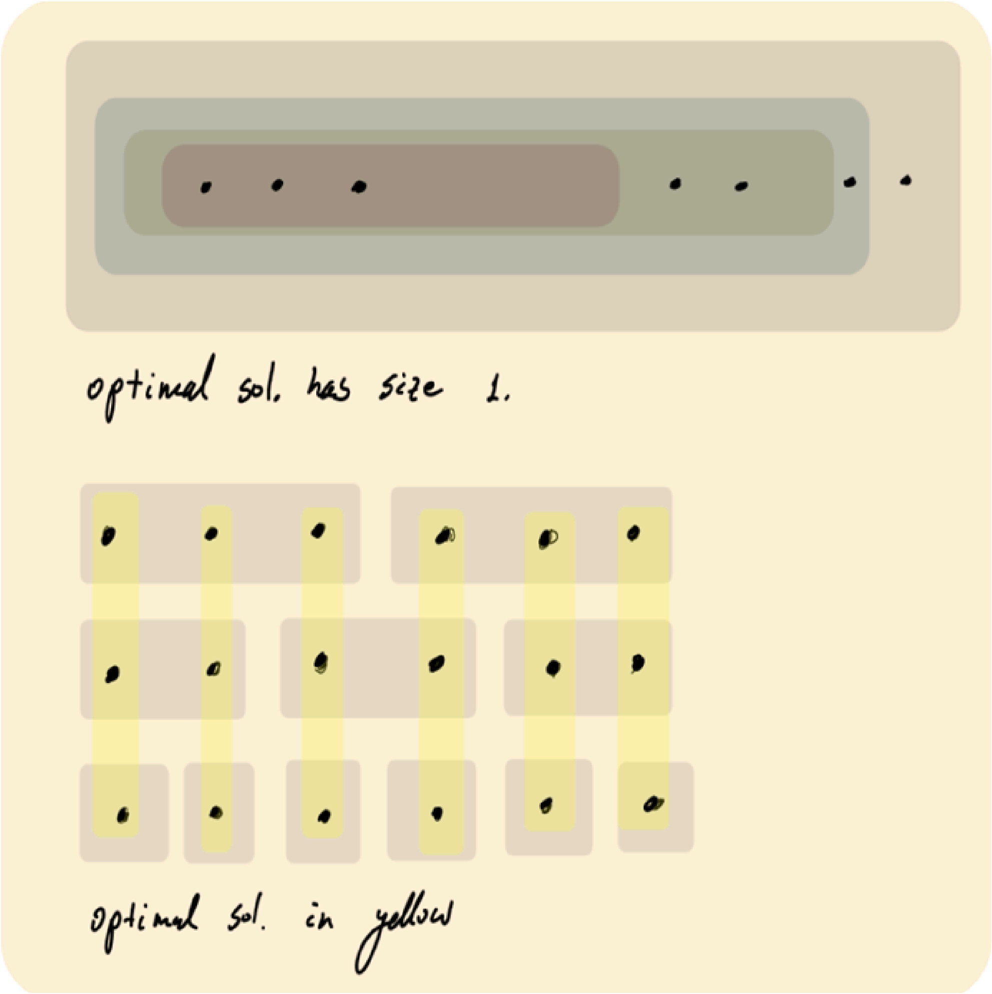
Instance: A hypergraph $H := (V, \mathcal{E})$.

Goal: A subset $C \subseteq \mathcal{E}$ such that $\bigcup_{e \in C} e = V$, minimising $|C|$.

Put another way: We seek a minimum **hyper-edge-cover** of H.

- Recall: for graphs, the minimum **edge-cover** problem is in P. **(Find maximum matching and extend greedily)**
- The minimum SC problem is **NP-complete** (we return to this below)

1.3 Some Examples



1.4 Min SC is NPC

$$\underbrace{\mathrm{VC} := \{(G,k) : \tau(G) \le k\}}_{\text{min vertex cover problem}} \qquad \mathrm{SC} := \{(H,k) : H \text{ has a hyper-edge cover of size} \le k\}$$

Claim: $\mathrm{VC} \le_p \mathrm{SC}$.

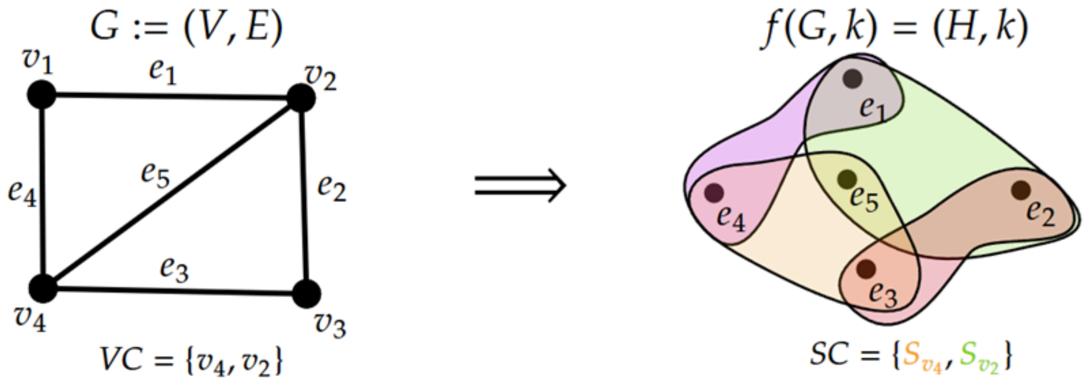
$\underbrace{\text{VC} := \{(G,k) : \tau(G) \le k\}}_{\text{min vertex cover problem}}$ SC $:= \{(H,k) : H$ has a hyper-edge cover of size $\le k\}$

Claim: VC $\le_p$ SC.

For each $u \in V(G)$, define: $S_u := \{e \in E(G) : u \in e\}$.

Given (G,n), define a hypergraph H by:

$$V(H) := E(G), \quad \mathcal{E}(H) := \{S_u : u \in V(G)\}$$



$$\underbrace{\mathrm{VC} := \{(G,k) : \tau(G) \le k\}}_{\text{min vertex cover problem}} \qquad \mathrm{SC} := \{(H,k) : H \text{ has a hyper-edge cover of size} \le k\}$$

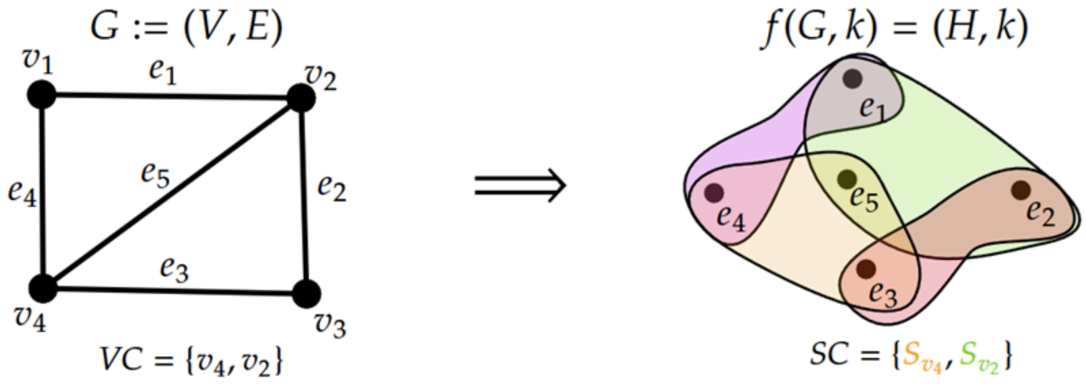
Claim: VC $\le_p$ SC.

For each $u \in V(G)$, define: $S_u := \{e \in E(G) : u \in e\}$.

Given (G, n), define a hypergraph H by:

$$V(H) := E(G), \quad \mathcal{E}(H) := \{S_u : u \in V(G)\}$$

Question 3 H.W. Complete the argument.



2. Greedy Algorithm for Minimum Set-Cover

Notation: For a set-system A, write $\bigcup A := \bigcup_{e \in A} e$.

Algorithm 1: Greedy-SC

```
A ← ∅
while ⋃A ≠ V do
    e ← arg max{|e \ ⋃A| : e ∈ ℰ}        (hyper-edge covering the most new vxs)
    A ← A ∪ {e}
end
return A
```

Notation: For a set-system A, write $\bigcup A := \bigcup_{e \in A} e$.

Algorithm 2: Greedy-SC

$A \leftarrow \emptyset$
while $\bigcup A \neq V$ **do**
  $e \leftarrow \arg\max\{|e \setminus \bigcup A| : e \in \mathcal{E}\}$ **(hyper-edge covering the most new vxs)**
  $A \leftarrow A \cup \{e\}$
end
return A

Question 5 What is the approximation ratio of the greedy algorithm?

2.2 Analysis Setup

Set $m := \max\{|e| : e \in \mathcal{E}\}$ – the size of the **largest** hyperedge.

Trivially $m \leq |V(H)| = n$**.**

2.2 Analysis Setup

Set $m := \max\{|e| : e \in \mathcal{E}\}$ – the size of the **largest** hyperedge.

Trivially $m \leq |V(H)| = n$.

Theorem 2 The greedy algorithm is an $O(\ln m)$-approximation for minimum set-cover.

2.2 Analysis Setup

Set $m := \max\{|e| : e \in \mathcal{E}\}$ – the size of the **largest** hyperedge.

Trivially $m \leq |V(H)| = n$.

Theorem 3 The greedy algorithm is an $O(\ln m)$-approximation for minimum set-cover.

Observation 8 Size of the cover output $=$ # iterations greedy performs.

Define: A_i = greedy solution at the **beginning** of the i-th iteration.

- Greedy transitions $A_i \to A_{i+1}$ during iteration i.
- # new vxs covered $= |\bigcup A_{i+1}| - |\bigcup A_i|$. **(all from $V(H) \setminus \bigcup A_i$)**

Let OPT = size of an optimal cover.

Claim : An optimal solution $P := \{e_1, ..., e_{\mathrm{OPT}}\}$ contains e_j s.t.:

$$\left|e_j \cap \left(V(H) \setminus \bigcup A_i\right)\right| \geq \frac{\left|V(H) \setminus \bigcup A_i\right|}{\mathrm{OPT}}$$

Proof (By contradiction):

- Assume no such e_j exists

$$\Longrightarrow \forall e \in \mathcal{E} : \left|e_j \cap \left(V(H) \setminus \bigcup A_i\right)\right| < \frac{\left|V(H) \setminus \bigcup A_i\right|}{\mathrm{OPT}}$$

Then,

$$\underbrace{|V(H)| = \left|\bigcup P\right|}_{P \text{ cover all of H}} < \mathrm{OPT} \cdot \frac{\left|V(H) \setminus \bigcup A_i\right|}{\mathrm{OPT}} = \underbrace{\left|V(H) \setminus \bigcup A_i\right| \leq |V(H)|}_{(V(H) \setminus \bigcup A_i) \subseteq V(H)}.$$

2.4 Key Observation

Since A_i is constructed **greedily**:

Observation 9 $\forall i \in \mathbb{N}: \quad |\bigcup A_{i+1}| - |\bigcup A_i| \geq \frac{|V(H) \setminus \bigcup A_i|}{\mathrm{OPT}}$

2.4 Key Observation

Since A_i is constructed **greedily**:

Observation 11 $\forall i \in \mathbb{N}: \quad |\bigcup A_{i+1}| - |\bigcup A_i| \geq \frac{|V(H) \setminus \bigcup A_i|}{\mathrm{OPT}}$

Q: Once A_{i+1} is defined, how many vxs are left to cover?

$$\begin{aligned} |V(H) \setminus \bigcup A_{i+1}| &= |V(H)| - |\bigcup A_{i+1}| \\ &\leq |V(H)| - |\bigcup A_i| - \frac{|V(H) \setminus \bigcup A_i|}{\mathrm{OPT}} \\ &= \left(1 - \frac{1}{\mathrm{OPT}}\right) |V(H) \setminus \bigcup A_i| \end{aligned}$$

2.4 Key Observation

Since A_i is constructed **greedily**:

Observation 13 $\forall i \in \mathbb{N}: \quad |\bigcup A_{i+1}| - |\bigcup A_i| \geq \frac{|V(H)\setminus\bigcup A_i|}{\mathrm{OPT}}$

Q: Once A_{i+1} is defined, how many vxs are left to cover?

$$\begin{aligned} |V(H) \setminus \bigcup A_{i+1}| &= |V(H)| - |\bigcup A_{i+1}| \\ &\leq |V(H)| - |\bigcup A_i| - \frac{|V(H) \setminus \bigcup A_i|}{\mathrm{OPT}} \\ &= \left(1 - \frac{1}{\mathrm{OPT}}\right) |V(H) \setminus \bigcup A_i| \end{aligned}$$

Observation 14 $|V(H) \setminus \bigcup A_{i+1}| \leq \left(1 - \frac{1}{\mathrm{OPT}}\right) |V(H) \setminus \bigcup A_i|$

After each iteration, the # of left-over vxs reduces by a factor of $(1 - 1/\mathrm{OPT})$.

2.5 Recursive Pattern

- Set $U_i := V(H) \setminus \bigcup A_i$ **(left-over vxs at iter i)**
- Intially $U_0 = V(H)$:

$$|U_{i+1}| \leq \left(1 - \frac{1}{\mathrm{OPT}}\right) |U_i|$$

By induction: $\forall k \geq 0$:

$$|U_k| \leq \left(1 - \frac{1}{\mathrm{OPT}}\right)^k \cdot |U_0| \leq \exp\left(-\frac{k}{\mathrm{OPT}}\right) \cdot n$$

- Set $U_i := V(H) \setminus \bigcup A_i$ **(left-over vxs at iter i)**
- Intially $U_0 = V(H)$:

$$|U_{i+1}| \leq \left(1 - \frac{1}{\mathrm{OPT}}\right) |U_i|$$

By induction: $\forall k \geq 0$:

$$|U_k| \leq \left(1 - \frac{1}{\mathrm{OPT}}\right)^k \cdot |U_0| \leq \exp\left(-\frac{k}{\mathrm{OPT}}\right) \cdot n$$

$$\exp\left(-\frac{i}{OPT}\right) \cdot n < 1$$
$$\ln\left(\exp\left(-\frac{i}{OPT}\right) \cdot n\right) < \ln(1)$$
$$-\frac{i}{OPT} + n < 0$$
$$n < \frac{i}{OPT}$$

The sequence $U_0, U_1, \ldots$ reaches zero when $\exp(-\frac{i}{\mathrm{OPT}}) \cdot n < 1$.

i.e. $$i > \mathrm{OPT} \cdot \ln n.$$

After $\mathrm{OPT} \cdot \ln n$ iterations, all of $V(H)$ is covered.

2.5 Recursive Pattern

- Set $U_i := V(H) \setminus \bigcup A_i$ **(left-over vxs at iter i)**
- Intially $U_0 = V(H)$:

$$|U_{i+1}| \leq \left(1 - \frac{1}{\mathrm{OPT}}\right) |U_i|$$

By induction: $\forall k \geq 0$:

$$|U_k| \leq \left(1 - \frac{1}{\mathrm{OPT}}\right)^k \cdot |U_0| \leq \exp\left(-\frac{k}{\mathrm{OPT}}\right) \cdot n$$

$$\exp\left(-\frac{i}{OPT}\right) \cdot n < 1$$

$$\ln\left(\exp\left(-\frac{i}{OPT}\right) \cdot n\right) < \ln(1)$$

$$-\frac{i}{OPT} + n < 0$$

$$n < \frac{i}{OPT}$$

The sequence $U_0, U_1, \ldots$ reaches zero when $\exp\left(-\frac{i}{\mathrm{OPT}}\right) \cdot n < 1$.

i.e. $$i > \mathrm{OPT} \cdot \ln n.$$

After $\mathrm{OPT} \cdot \ln n$ iterations, all of $V(H)$ is covered.

Recalling that size of greedy sol = # iterations:

Theorem 6 size of greedy sol $\leq (\ln n) \cdot \mathrm{OPT}$ — an $O(\ln n)$-approximation.

2.6 Tightening to $O(\ln m)$

Subtlety: The theorem asserts $O(\ln m)$ but we got $O(\ln n)$. Since n and m can differ greatly, we refine.

2.6 Tightening to $O(\ln m)$

Subtlety: The theorem asserts $O(\ln m)$ but we got $O(\ln n)$. Since n and m can differ greatly, we refine.

Choose i^* as the last iteration where $\mathrm{OPT} \leq |U_{i^*}|$.

- After i^* iters: $|U_{i^*+1}| < \mathrm{OPT}$. **(Cover the rest using at most $\mathrm{OPT} - 1$ hyper-edges)**
- So: size of greedy sol $\leq (i^* + 1) + (\mathrm{OPT} - 1) = i^* + \mathrm{OPT}$.

2.6 Tightening to $O(\ln m)$

Subtlety: The theorem asserts $O(\ln m)$ but we got $O(\ln n)$. Since n and m can differ greatly, we refine.

Choose i^* as the last iteration where $\text{OPT} \leq |U_{i^*}|$.

- After i^* iters: $|U_{i^*+1}| < \text{OPT}$. **(Cover the rest using at most $\text{OPT} - 1$ hyper-edges)**
- So: size of greedy sol $\leq (i^* + 1) + (\text{OPT} - 1) = i^* + \text{OPT}$.

Estimating i^*:

$$\text{OPT} \leq |U_{i^*}| \leq \left(1 - \frac{1}{\text{OPT}}\right)^{i^*} \cdot |U_0| \leq \exp(-i^*/\text{OPT}) \cdot n.$$

$$\exp(i^*/\ \text{OPT}) \leq \frac{n}{\text{OPT}} \quad \Longrightarrow \quad i^* \leq \text{OPT} \cdot \ln\left(\frac{n}{\text{OPT}}\right).$$

$$\exp\left(\frac{i^*}{OPT}\right) \leq \frac{n}{OPT}$$

$$\ln\left(\exp\left(\frac{i^*}{OPT}\right)\right) \leq \ln\left(\frac{n}{OPT}\right)$$

$$\frac{i^*}{OPT} \leq \ln\left(\frac{n}{OPT}\right)$$

$\Longrightarrow$ size of greedy sol $\leq \text{OPT} \cdot \ln\left(\frac{n}{\text{OPT}}\right) + \text{OPT} = \left(1 + \ln\left(\frac{n}{\text{OPT}}\right)\right) \cdot \text{OPT}$.

Subtlety: The theorem asserts $O(\ln m)$ but we got $O(\ln n)$. Since n and m can differ greatly, we refine.

Choose i^* as the last iteration where $\mathrm{OPT} \le |U_{i^*}|$.

- After i^* iters: $|U_{i^*+1}| < \mathrm{OPT}$. **(Cover the rest using at most $\mathrm{OPT} - 1$ hyper-edges)**
- So: size of greedy sol $\le (i^* + 1) + (\mathrm{OPT} - 1) = i^* + \mathrm{OPT}$.

Estimating i^*:

$$\mathrm{OPT} \le |U_{i^*}| \le \left(1 - \frac{1}{\mathrm{OPT}}\right)^{i^*} \cdot |U_0| \le \exp(-i^*/\mathrm{OPT}) \cdot n.$$

$$\exp(i^*/\,\mathrm{OPT}) \le \frac{n}{\mathrm{OPT}} \implies i^* \le \mathrm{OPT} \cdot \ln\left(\frac{n}{\mathrm{OPT}}\right).$$

$$\exp\left(\frac{i^*}{OPT}\right) \le \frac{n}{OPT}$$
$$\ln\left(\exp\left(\frac{i^*}{OPT}\right)\right) \le \ln\left(\frac{n}{OPT}\right)$$
$$\frac{i^*}{OPT} \le \ln\left(\frac{n}{OPT}\right)$$

$\implies$ size of greedy sol $\le \mathrm{OPT} \cdot \ln(\frac{n}{\mathrm{OPT}}) + \mathrm{OPT} = (1 + \ln(\frac{n}{\mathrm{OPT}})) \cdot \mathrm{OPT}$.

Observation 18 $m = \max\{|e| : e \in \mathcal{E}\} \ge \frac{n}{\mathrm{OPT}}$ since every optimal solution must contain a hyperedge of size $\ge n/\mathrm{OPT}$.

$$\text{size of greedy sol} \le (1 + \ln m) \cdot \mathrm{OPT} = O(\ln m) \cdot \mathrm{OPT}. \quad \square$$

3. Minimum Cost Set-Cover

Problem 6 Minimum Cost Set-Cover

Instance: A hypergraph H; a cost function $c : \mathcal{E}(H) \to \mathbb{R}_{\geq 0}$.

Goal: An edge-cover $C \subseteq \mathcal{E}(H)$ of H minimising $\sum_{e \in C} c(e)$.

3.1 The Weighted Problem

Problem 7 Minimum Cost Set-Cover

Instance: A hypergraph H; a cost function $c : \mathcal{E}(H) \to \mathbb{R}_{\geq 0}$.

Goal: An edge-cover $C \subseteq \mathcal{E}(H)$ of H minimising $\sum_{e \in C} c(e)$.

The cardinality greedy **fails** for the cost/weight version. Can we still learn from it?

3.2 Reformulating the Greedy Choice

Consider the i-th iteration (with $V(H) \setminus \bigcup A_i \neq \emptyset$).

The cardinality greedy picks $e \in \mathcal{E}(H) \setminus A_i$ maximising:

$$\frac{|e \setminus \bigcup A_i|}{|V(H) \setminus \bigcup A_i|} = \frac{\#\text{ new vxs covered by } e}{\#\text{ left-over vxs}}$$

3.2 Reformulating the Greedy Choice

Consider the i-th iteration (with $V(H) \setminus \bigcup A_i \neq \emptyset$).

The cardinality greedy picks $e \in \mathcal{E}(H) \setminus A_i$ maximising:

$$\frac{|e \setminus \bigcup A_i|}{|V(H) \setminus \bigcup A_i|} = \frac{\#\text{ new vxs covered by } e}{\#\text{ left-over vxs}}$$

For weighted: we would want an edge $e \in \mathcal{E}(H) \setminus A_i$ maximising:

$$\frac{\text{"cost" of new vxs covered by } e}{\text{"cost" of vxs left-over}}$$

(not well-defined)

3.2 Reformulating the Greedy Choice

Consider the i-th iteration (with $V(H) \setminus \bigcup A_i \neq \emptyset$).

The cardinality greedy picks $e \in \mathcal{E}(H) \setminus A_i$ maximising:

$$\frac{|e \setminus \bigcup A_i|}{|V(H) \setminus \bigcup A_i|} = \frac{\#\text{ new vxs covered by } e}{\#\text{ left-over vxs}}$$

For weighted: we would want an edge $e \in \mathcal{E}(H) \setminus A_i$ maximising:

$$\frac{\text{"cost" of new vxs covered by } e}{\text{"cost" of vxs left-over}}$$

(not well-defined)

Back to cardinality: For $e \in \mathcal{E}(H) \setminus A_i$, consider $1/\ |e \setminus \bigcup A_i|$.

- The **smaller** this ratio, the **more** new elements e covers.

For weighted: $c(e)/\ |e \setminus \bigcup A_i|$ – distribute cost $c(e)$ among newly covered vxs. **(This is amortisation!)**
Pick $e \in \mathcal{E}(H) \setminus A_i$ minimising

$$\frac{c(e)}{|e \setminus \bigcup A_i|} = \frac{\text{cost of } e}{\#\text{ vxs covered by } e}$$

3.3 Weighted Greedy Algorithm

Definition 8 Effectiveness

Let $U \subseteq V(H)$ and $e \in \mathcal{E}(H)$. Define:

$$\mathrm{eff}_U(e) := \begin{cases} \infty & \text{if } e \subseteq U \\ \dfrac{c(e)}{|e \setminus U|} & \text{otherwise} \end{cases}$$

to be the **effectiveness** of e with respect to U.

Algorithm 3: Greedy-Cost-SC

```
1: A ← ∅
2: while ⋃A ≠ V do
3:     e ← arg min{ eff_{⋃A}(e) : e ∈ ℰ(H) }
4:     A ← A ∪ {e}
5: end
6: return A
```

Definition 9 Effectiveness

Let $U \subseteq V(H)$ and $e \in \mathcal{E}(H)$. Define:

$$\mathrm{eff}_U(e) := \begin{cases} \infty & \text{if } e \subseteq U \\ \dfrac{c(e)}{|e \setminus U|} & \text{otherwise} \end{cases}$$

to be the **effectiveness** of e with respect to U.

Algorithm 4: Greedy-Cost-SC

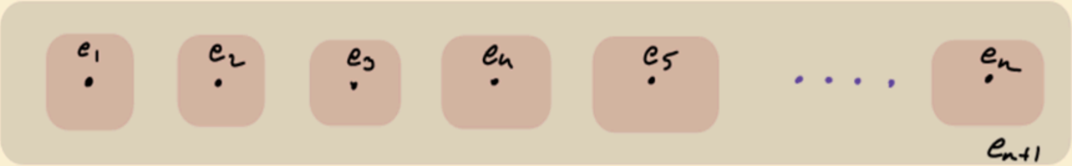
1: $A \leftarrow \emptyset$
2: **while** $\bigcup A \neq V$ **do**
3: $\quad e \leftarrow \arg\min\left\{\mathrm{eff}_{\bigcup A}(e) : e \in \mathcal{E}(H)\right\}$
4: $\quad A \leftarrow A \cup \{e\}$
5: **end**
6: **return** A

Theorem 8 The greedy algorithm above forms an $O(\ln n)$-approximation for minimum cost set-cover.

3.4 Example: Greedy vs Optimal

- An example for how the algo "works"

$H :=$



$c(e_1) = \frac{1}{n}$ $\quad c(e_2) = \frac{1}{n-1}$ $\quad c(e_3) = \frac{1}{n-2}$ $\quad \dots \quad c(e_n) = \frac{1}{n-(n-1)} = 1$

$c(e_{n+1}) = 1+\varepsilon$, $\quad \varepsilon > 0$ predetermined

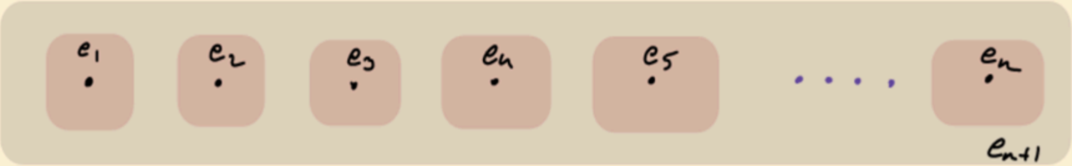
OPTIMAL SOLUTION:

Take e_{n+1} only with cost $1+\varepsilon$

3.4 Example: Greedy vs Optimal

An example for how the algo "works"

$U :=$



$c(e_1) = \frac{1}{n}$ $\quad c(e_2) = \frac{1}{n-1}$ $\quad c(e_3) = \frac{1}{n-2}$ $\quad \ldots\ldots \quad c(e_n) = \frac{1}{n-(n-1)} = 1$

$c(e_{n+1}) = 1 + \varepsilon$, $\varepsilon > 0$ predetermined

OPTIMAL SOLUTION:

Take e_{n+1} only with cost $1+\varepsilon$

Greedy (step 1, $A = \emptyset$):

- $\text{eff}_{\emptyset(e_i)} = c(e_i)/1.$
- $\text{eff}_{\emptyset(e_{n+1})} = (1+\varepsilon)/n.$

$\Longrightarrow e_1$ chosen. **(eff $= 1/n$, the smallest)**

Greedy (step 2, $A = \{e_1\}$):

- $\text{eff}_{\{e_1\}}(e_i) = c(e_i)/1$
- $\text{eff}_{\{e_1\}}(e_{n+1}) = (1+\varepsilon)/(n-1)$

$\Longrightarrow e_2$ chosen. **(eff $= \frac{1}{n-1}$, the smallest)**

...

Greedy selects $e_1, e_2, \ldots, e_n$ with total cost:

$$\sum_{i=1}^{n} c(e_i) = \sum_{i=1}^{n} \frac{1}{n+1-i} = \sum_{j=1}^{n} \frac{1}{j} \approx \ln n \quad (\text{OPT} = 1 + \varepsilon)$$

4. Analysis of Weighted Greedy

4.1 Basic Intuition

Claim : Let P be an optimal cover with cost OPT. Then $\exists e \in P$ such that $c(e)/|e| \leq \mathrm{OPT}/n$.

***Proof*.** Suppose $c(e)/|e| > \mathrm{OPT}/n$ for all $e \in P$. Then:

$$\mathrm{OPT} = \sum_{e \in P} c(e) > \sum_{e \in P} |e| \cdot \frac{\mathrm{OPT}}{n} = \underbrace{\frac{\mathrm{OPT}}{n} \cdot \sum_{e \in P} |e| \geq \frac{\mathrm{OPT}}{n} \cdot n}_{\substack{\sum_{e \in P} |e| \geq |\bigcup_{e \in P} e| = n \\ P \text{ covers all } n \text{ vxs.}}} = \mathrm{OPT}. \quad .$$

$\square$

Claim : Let P be an optimal cover with cost OPT. Then $\exists e \in P$ such that $c(e)/|e| \leq \mathrm{OPT}/n$.

Proof. Suppose $c(e)/|e| > \mathrm{OPT}/n$ for all $e \in P$. Then:

$$\mathrm{OPT} = \sum_{e \in P} c(e) > \sum_{e \in P} |e| \cdot \frac{\mathrm{OPT}}{n} = \underbrace{\frac{\mathrm{OPT}}{n} \cdot \sum_{e \in P} |e| \geq \frac{\mathrm{OPT}}{n} \cdot n}_{\substack{\sum_{e \in P} |e| \geq |\bigcup_{e \in P} e| = n \\ P \text{ covers all } n \text{ vxs.}}} = \mathrm{OPT}. \quad .$$

□

Amortised perspective:

Observation 20 $\exists e \in P$ covering its vxs at amortised cost $\leq \mathrm{OPT}/n$ **per vertex**.

Observation 21 $\exists e \in P$ covering its vxs at amortised cost $\leq \mathrm{OPT}/n$ **per vertex**.

This argument persists at **later stages**. Let P = optimal cover, A_i = current greedy cover. Define:

$$T := \{e \in P : e \setminus \bigcup A_i \neq \emptyset\}$$

(All hyper-edges in P that cover "new" vertices)

- Only members of T are relevant to the greedy: $\mathrm{eff}_{\bigcup A_i}(e) < \infty$ iff $e \in T$
- $(V(H) \setminus \bigcup A_i) \subseteq \bigcup T$

(exercise: proof by contradiction, i.e. remaining vertices are covered by T.)

Lemma 4.2.1 (Key Claim) $\exists e \in T$ s.t.

$$\mathrm{eff}_{\bigcup A_i}(e) = \frac{c(e)}{|e \setminus \bigcup A_i|} \leq \frac{\mathrm{OPT}}{|V(H) \setminus \bigcup A_i|}.$$

Lemma 4.2.2 (Key Claim) $\exists e \in T$ s.t.

$$\mathrm{eff}_{\bigcup A_i}(e) = \frac{c(e)}{|e \setminus \bigcup A_i|} \leq \frac{\mathrm{OPT}}{|V(H) \setminus \bigcup A_i|}.$$

Proof. Suppose not. Then:

$$\begin{aligned}\mathrm{OPT} \geq \sum_{e \in T} c(e) &> \frac{\mathrm{OPT}}{|V(H) \setminus \bigcup A_i|} \cdot \sum_{e \in T} |e \setminus \bigcup A_i| \\ &\underbrace{\geq \frac{\mathrm{OPT}}{|V(H) \setminus \bigcup A_i|} \cdot |V(H) \setminus \bigcup A_i|}_{\sum_{e \in T} |e \setminus \bigcup A_i| \geq |\bigcup (T \setminus \bigcup A_i)| = |V(H) \setminus \bigcup A_i|} = \mathrm{OPT}. \quad \square\end{aligned}$$

4.3 Amortised Cost Tracking

Augment the greedy to assign a **price** to each vertex when first covered:

Algorithm 5: Greedy-Cost-SC (augmented)

```
A ← ∅
while ⋃A ≠ V do
    e ← arg min{eff_{⋃A}(e) : e ∈ 𝓔(H)}
    for each v ∈ e \ ⋃A do
        Set α(v) := eff_{⋃A}(e)
    end
    A ← A ∪ {e}
end
return A
```

Reformulation: Choose $e \in \mathcal{E}(H)$ whose avg cost per **newly covered** element is least.

Lemma 4.4.1 (Cost) cost of greedy sol $= \sum_{v \in V} \alpha(v)$.

Proof. Let $A = \{e_1, ..., e_k\}$ be the greedy sol; $A_j = \{e_1, ..., e_j\}$ the partial sol at step j.

$$\sum_{e \in A} c(e) = \sum_{j=1}^{k} c(e_j) = \sum_{j=1}^{k} \sum_{v \in e_j \setminus \bigcup A_j} \underbrace{\frac{c(e_j)}{|e_j \setminus \bigcup A_j|}}_{=\alpha(v)} = \sum_{v \in V} \alpha(v). \quad \square$$

Theorem 12 The weighted greedy algorithm is an $O(\ln n)$-approximation.

Proof. Order $V = \{v_1, v_2, ..., v_n\}$ by the order the greedy covers them.

When v_i is covered at iteration j, there are $\geq n - i + 1$ uncovered vxs. By **Lemma 4.2.2**:

$$\alpha(v_i) = \mathrm{eff}_{\bigcup A_j}(e_j) \leq \frac{\mathrm{OPT}}{|V(H) \setminus \bigcup A_j|} \leq \frac{\mathrm{OPT}}{n - i + 1}.$$

4.5 Approximation Ratio

Theorem 13 The weighted greedy algorithm is an $O(\ln n)$-approximation.

Proof. Order $V = \{v_1, v_2, \ldots, v_n\}$ by the order the greedy covers them.

When v_i is covered at iteration j, there are $\geq n - i + 1$ uncovered vxs. By **Lemma 4.2.2**:

$$\alpha(v_i) = \mathrm{eff}_{\bigcup A_j}(e_j) \leq \frac{\mathrm{OPT}}{|V(H) \setminus \bigcup A_j|} \leq \frac{\mathrm{OPT}}{n - i + 1}.$$

Therefore, using **Lemma 4.4.1**

$$\text{cost of greedy sol} = \sum_{v \in V} \alpha(v) = \sum_{i=1}^{n} \alpha(v_i) \leq \mathrm{OPT} \cdot \sum_{i=1}^{n} \frac{1}{n - i + 1} = \mathrm{OPT} \cdot H_n.$$

where $H_n = \sum_{i=1}^{n} 1/i \approx \ln n$ is the n-th **Harmonic number**. $\square$